

Global Nickel Short-Term Outlook

December 2024



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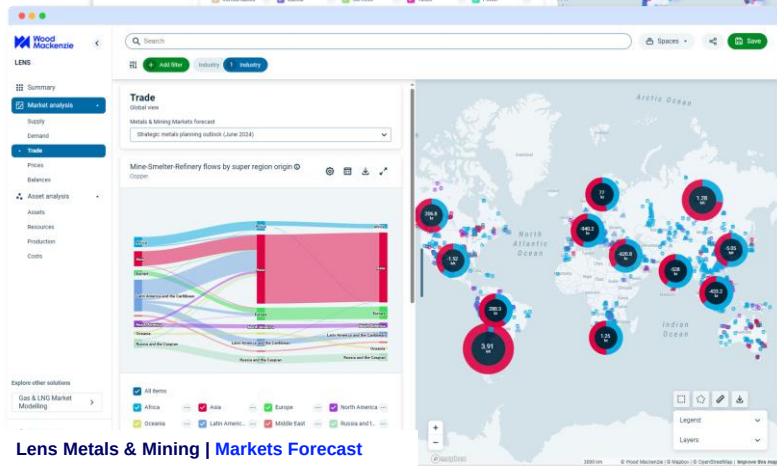
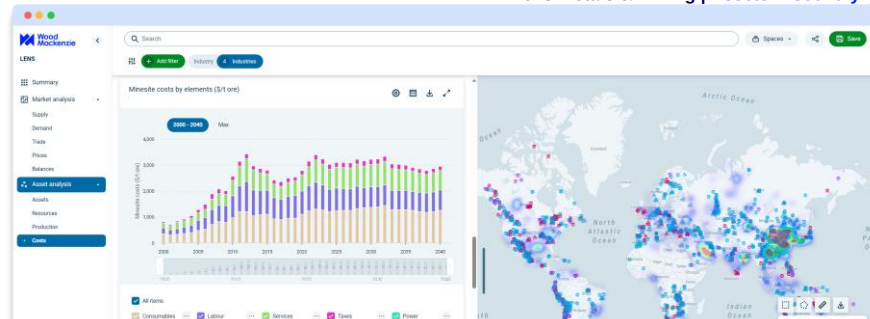
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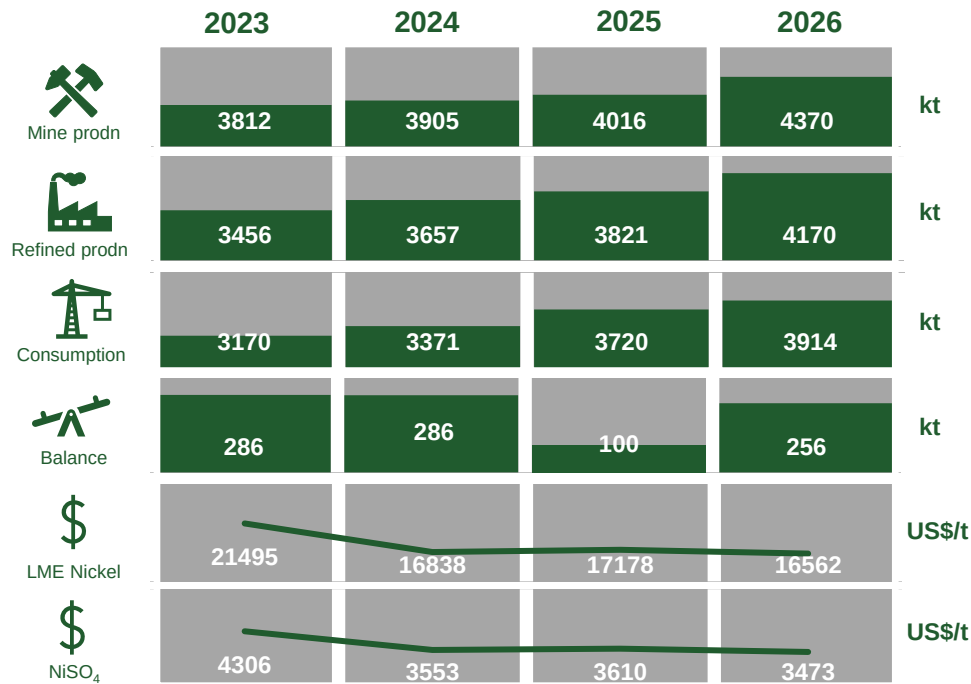
Executive summary

Nickel key forecasts

Nickel markets are fading out of 2024 with a whimper. Prices remain entrenched in the US\$15,500-15,800/t range (US\$7.03-7.17/lb) and nothing this side of the month's end is apparent which will shake it out of this quagmire. Global demand growth of 6.4% would ordinarily have markets jumping for joy and prices reaching new ceilings. But supply side growth has been stronger still, leaving the market overflowing with product irrespective of its form or purpose.

Can nickel turn its fortunes around as we enter the New Year? There are glimmers of hope. We saw in 2024 significant cutbacks in supply (despite what will still be 6% growth on 2023), and the potential for further cuts in New Caledonia (Doniambo) and Indonesia (lower mine feed) could jolt the market back to life early on in Q1. Equally, the Chinese New Year celebrations come early – at the end of January – and so the world will be anticipating demand growth as soon as February, especially if any elements of the recent stimulus package feed through. From physical demand at home to easing the paths for foreign investor trading, Beijing has its eye on putting its economy on an upward trajectory in 2025.

For nickel, we have the chance to see stronger stainless markets in Europe and China at least. But the battery sector needs to counter the advances made by lithium iron phosphate. Entering 2025 with a bang might not be the best choice of phrase when it comes to nickel and batteries. But there is a clear need to regain momentum.



Source: Wood Mackenzie

Nickel balances by type

Class I – tight supply in the RoW in 2021-2022 supported nickel prices. Building oversupply from 2023 as China started delivering metal to LME warehouses, especially from 2024.

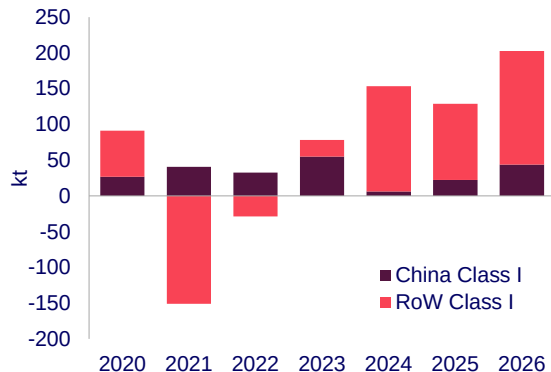
Class II – burgeoning Indonesian NPI production and sustained imports of western FeNi led to oversupply in China in 2022-2023. Closure of many western FeNi producers and stronger stainless production has tightened supply in 2024. Market will be even tighter in 2025-2026.

Nickel sulphate - balanced market has moved into oversupply in 2024 as Chinese battery demand has not kept pace with supply, itself boosted by expansion in supply of intermediates in Indonesia. Some of the excess has been further processed to Class I, which has gone to the exchanges, and this practise will continue in 2025.

	2020	2021	2022	2023	2024	2025	2026
Class I							
China Class I	27	40	32	55	6	22	44
RoW Class I	65	-151	-29	23	147	107	159
Total Class I	91	-111	3	78	153	129	203
Class II							
China Class II	16	0	163	202	1	-97	-58
RoW Class II	18	17	16	11	73	0	0
Total Class II	35	16	179	213	74	-97	-58
Nickel sulphate							
China sulphate	-27	15	-5	13	54	47	70
RoW sulphate	-6	-29	-27	-18	4	21	42
Total sulphate	-33	-14	-33	-5	58	68	111
Total nickel	93	-108	150	286	286	100	256

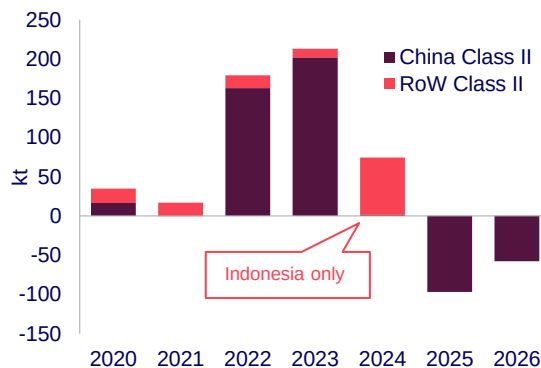
Source: Wood Mackenzie

Class I nickel balance



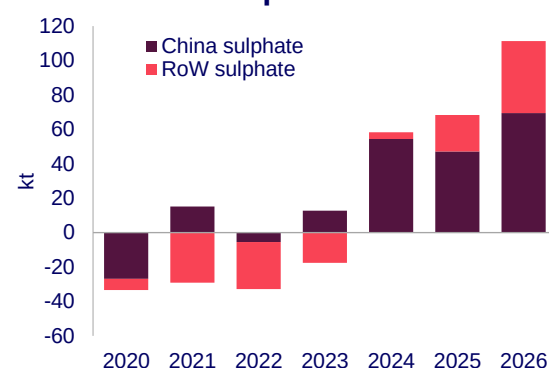
Source: Wood Mackenzie

Class II nickel balance



Source: Wood Mackenzie

Nickel sulphate balance



Source: Wood Mackenzie

Prices

Nickel finds its floor

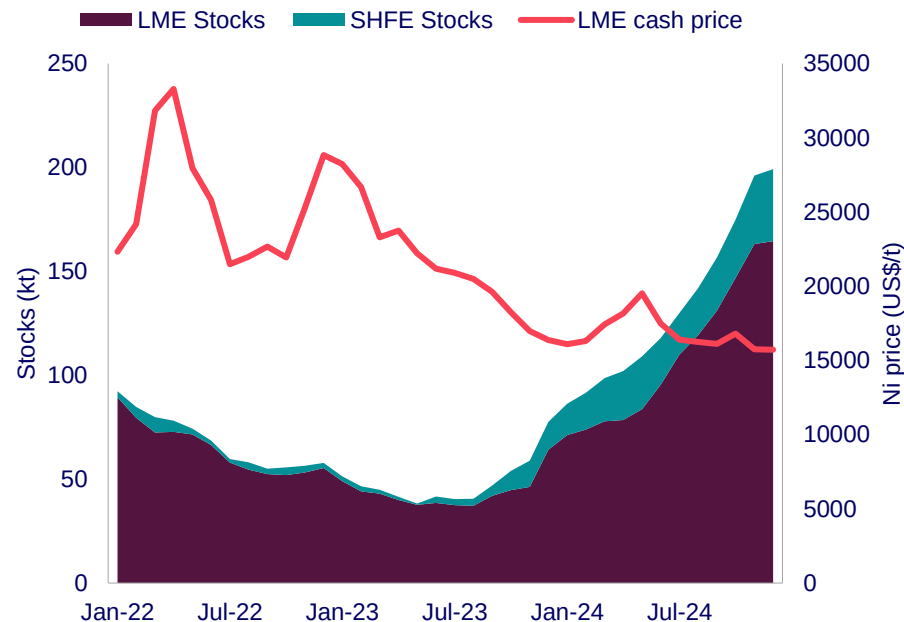
Supply-side action has stalled further price decline – now we need to better manage demand expectations

Despite flirting with the US\$16,000/t (US\$7.25/lb) intraday price target on 12 December, nickel bears kept the metal on a par with November's four-year low of US\$15,740/t (US\$7.14/lb) until well into the third trading week of December. With little excitement on the horizon for nickel and as the holidays approach, we expect the final quarter's price to average around US\$16,000/t, leaving the 2024 average at US\$16,840/t (US\$7.64/lb), or US\$4,700/t (US\$2.13/lb) lower than 2023.

In a year which has seen persistently disappointing market commentary on nickel's two principal areas of demand – stainless and batteries - and hefty increments in visible stocks on both the world's main exchanges, the flagging nickel price comes as no surprise. Add to that a well-advertised 11.5% jump in Class II nickel pig iron (NPI) output from Indonesia this year and one begins to appreciate the pressure that nickel prices have come under.

However, it is positive that prices have not crashed further. Reasons why include the fact that global demand has risen by 6.5% for a second consecutive year, suggesting that it is our expectations of a better performance that we are not managing well: that 68% of the 151.8 kt of open stocks in LME depots on 29 November were from sources that western customers are trying to avoid; and that several sources of supply – from Chinese NPI to western sulphides – have reacted to cost pressure and closed.

Exchange nickel stocks and prices



Source: LME; SHFE; Wood Mackenzie

Glass half empty, or half full? Trading opportunities will exist in 2025...

..but they'll be harder to find in 2026

What can we expect in 2025? The headline market surplus of +100 kt, following on from the +286 kt for each of 2023 and 2024 suggests “more of the same”. But we would prefer to say that market players will need to look harder into the details to find opportunities! Another 100 kt does sound unappealing, but it is a lot better than what the prior two years offered, especially if you consider that our forecast rate of demand growth next year (10.4%) is more than double that of supply. And even if this is over-optimistic for demand in 2025, there is still further upside from supply: mine feed inside Indonesia might be curtailed because of lower license issuance; and doubts remain over the future of SLN's Doniambo in New Caledonia. Both outcomes will have important influences in Q1 next year.

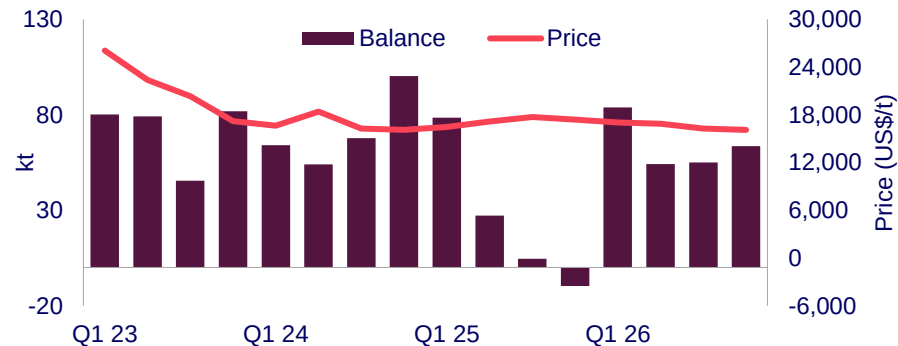
Another “opportunity”, we believe, lies in maintaining close scrutiny of LME stocks. As we said above, stock levels on the LME (and in Shanghai) keep rising, but that does not automatically represent an improvement in liquidity. Chinese, Indonesian and Russian metal is not the first source western consumers want to turn to from the market of last resort. But aside from fresh briquettes from Impala, following its recent re-instatement by the LME, there are few other choices available to consumers; the LME is not immune from the odd squeeze!

And that potential could build by mid-year. Declining visible surpluses during H1, plus what we expect to be a “tighter” market in the second half, will present trading opportunities which will keep metal prices buoyant (with spot highs potentially above US\$18,000/t).

But we expect those opportunities to vanish as we head into 2026, counting on the basis of current exchange products and services. As we will discuss in an insight to be published next month, the trading vehicles offered by the SHFE are likely to widen in scope (as they have already for other commodities already this year), and these collectively might change the nickel trading landscape beyond fair comparison with what market players can do today.

But looking at the market balance as it stands, we expect a return to substantial surplus and price pain in 2026. After averaging US\$17,200/t (US\$7.80/lb, current) in 2025, prices will slide again to below US\$16,600/t (US\$7.50/lb) in 2026.

Quarterly outlook for total nickel market balances and price



Source: Wood Mackenzie

Nickel sulphate supply/demand and price

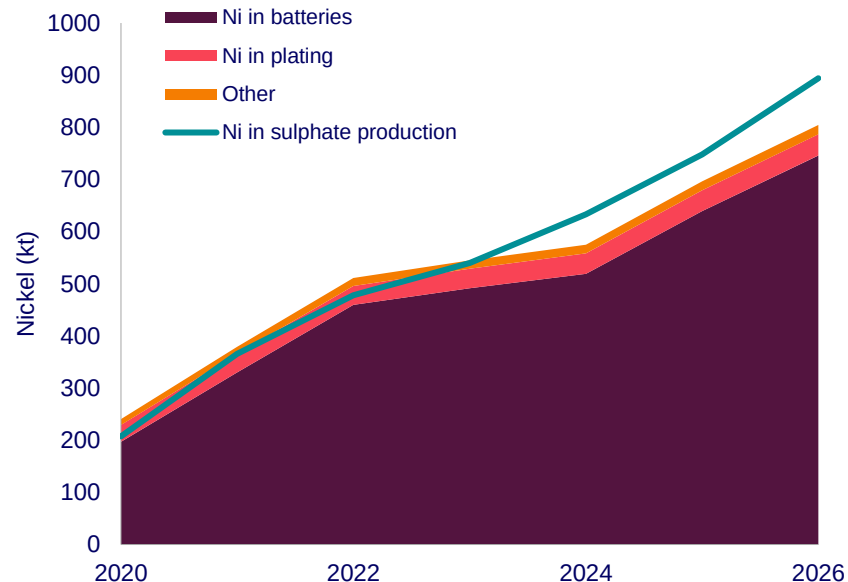
Premiums outside China are stable. The rise in Chinese prices from July ground to a halt in November

China: Chinese domestic nickel sulphate prices, having risen since July, fell sharply in November. That said, as suggested last month, discounts did narrow with the LME to average around US\$1,395/t nickel. Our average discount for the year for Chinese nickel in sulphate stands at US\$1,346/t, increasing in 2025 to US\$1,900/t.

RoW: Korean nickel sulphate was achieving a premium of around US\$1,500/t over October. Japanese premiums remained around US\$5,000/t for September. Overall, for 2024 we expect non-Chinese sulphate to be sold at an average premium of around US\$2,000/t nickel, and for this to decline in 2025 to around US\$1,900/t.

Indonesia: Chinese nickel sulphate imports from Indonesia were discounted more heavily than Chinese domestic material. For 2023 the discount averaged around US\$2,750/t. For 2024, data to October show the Indonesian discount averaged US\$3,100/t. Harita, the sole producer of sulphate in Indonesia, currently put its sulphate sales price for nine months of 2024 at US\$14,344/t Ni – an average discount of US\$2,740/t nickel.

Global NiSO₄ supply and demand



Source: Wood Mackenzie

Demand

Changes in forecast nickel demand this month

Minor revisions only

We have made small changes in both stainless and battery demand this month.

In stainless, global output is revised up by 130 kt in 2024 and 185 kt in 2025, most of that being attributable to China (based on the latest data). Together with small adjustments to estimated %Ni content in stainless, these changes equate to around 10 ktpa more Chinese nickel consumption in 2024-2025.

European stainless is also revised up by 40-60 ktpa for 2024-2026, with a negligible change in primary nickel uptake.

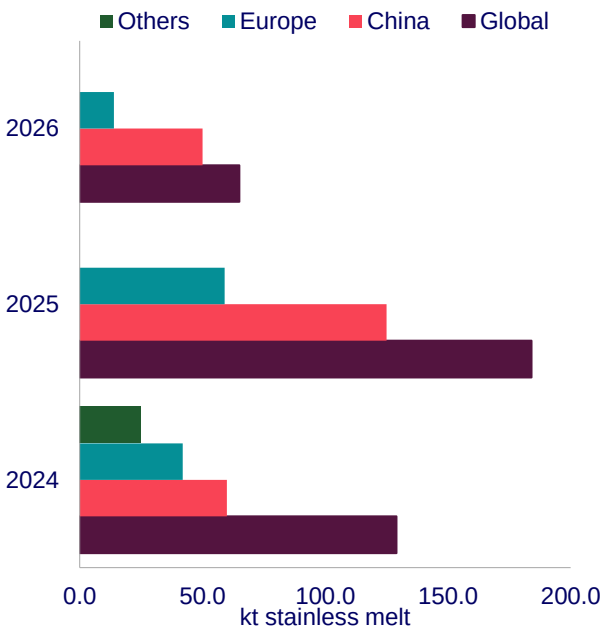
In pCAMs for batteries, nickel consumption was revised down in 2024 by 4 kt, and up in 2025 by 5 kt, largely because we have pushed back some forthcoming supply by a year.

Main changes in nickel consumption



Source: Wood Mackenzie

Main changes in stainless melt output



Source: Wood Mackenzie

Short-term outlook for stainless steel – global growth driven mostly by China

2025 is better, assuming global economies improve, but risks remain on the downside

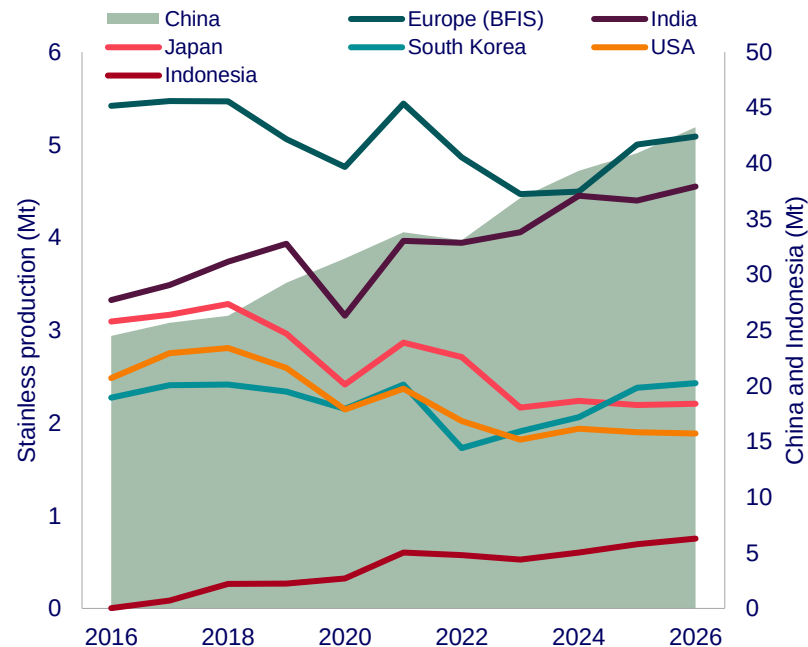
Global stainless production will increase by 6.4% in 2024, to 62.8 Mt, and continue growing at around 5% a year, to 69.4 Mt in 2026. Growth is primarily driven by China, which accounts for more than 60% of global output. However, we also forecast improvements in Europe (better economy and no strikes), Indonesia (through a rejuvenated Delong and new capacity from Jindal), and South Korea, where Posco finally returns to pre-typhoon utilisation.

European growth of 10%, to 6.8 Mt, may appear optimistic right now, but with both Acerinox (in Spain) and Outokumpu (in Finland) unaffected by strikes a substantial percentage increase in those countries is almost inevitable. Regionally, however, the forecast of 6.8 Mt in 2025 remains well below the 7.4-7.5 Mt of the last peak in 2021 and the pre-Covid average of 2011-2018.

Some European mills have imported Indonesian NPI this year and this practise will continue in 2025, for as long as scrap supply remains tight. Therefore, European primary nickel consumption will rise more steeply in 2025 than the increase in austenitic stainless alone may indicate.

Global primary nickel consumption in stainless will increase by 7% in 2024, to 2.18 Mt, and then by 9% in 2025, to 2.4 Mt.

Stainless steel production outlook by country



Source: Wood Mackenzie

BFIS=Belgium, Finland, Italy, Spain

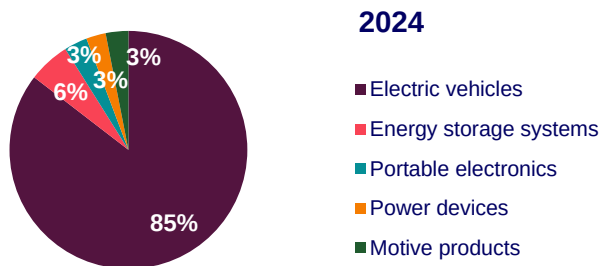
Short-term outlook for nickel in battery precursors (pCAMs)

Mounting risks to what has long been a positive outlook for nickel chemistries

The ongoing weakness in battery and battery materials markets continues. Global nickel consumption in pCAMs will increase by 5.6% in 2024, to 519 kt, before rising with a CAGR of 20% to 746 kt in 2026, most of this coming in China.

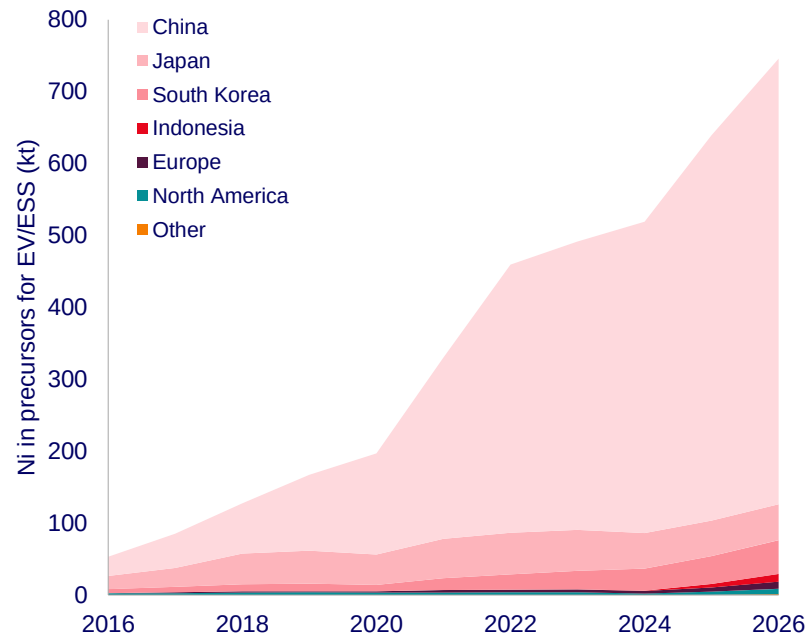
Japan and South Korea are the other two main pCAM producers. But while Japanese demand for nickel is steady at around 50 ktpa, new capacity in South Korea will boost its demand from 31 kt this year to about 47 kt in 2026. New Indonesian capacity will also start to ramp up in 2025.

However, risks remain on the downside, the main threat outside EV and battery demand being the preference for LFP (lithium iron phosphate) chemistry, which contains no nickel and could account for 50-60% of the battery market in the future.



Source: Wood Mackenzie

Global nickel consumption in pCAMs



Source: Wood Mackenzie

Short-term outlook for global nickel consumption

Asia, especially China, is driving growth due to regional dominance in both stainless steel and batteries

Global nickel consumption will increase by 6.4% in 2024, to 3.37 Mt, boosted by a substantial increase in demand in China, despite weakening stainless and battery markets. Battery markets, while better than in 2023, have steadied since mid-year.

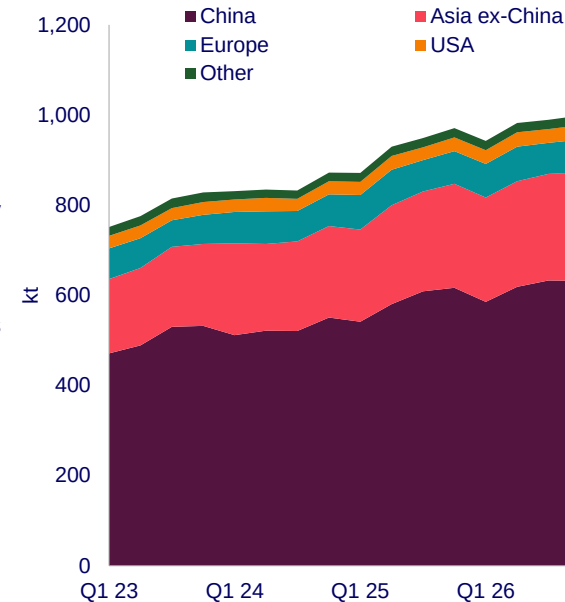
Elsewhere, increased stainless production supports stronger nickel demand in Indonesia and India, while use of Indonesian NPI in Europe has boosted primary nickel use there, and will do so again in 2025.

Growth of 10.4%, to 3.72 Mt in 2025, is mostly accounted for by China (helped by economic stimulus and a fall in stainless scrap use) and Indonesia, and also assumes some recovery in other regional economies.

Growth will slow to 5.2% in 2026, with global consumption rising to 3.91 Mt in that year.

Stainless remains the principal consumer of nickel although the share of global demand going to battery pCAMs is growing, from 16% in 2024 to 19% in 2026.

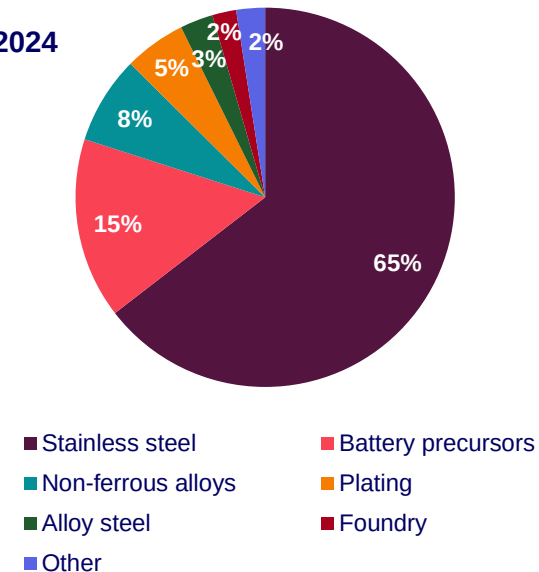
Refined nickel consumption by region



Source: Wood Mackenzie

Refined nickel consumption by first use

2024



Source: Wood Mackenzie

Chinese stainless production rising despite off-colour demand

Demand is not keeping up with supply and stainless prices are still falling – to a four-year low in December

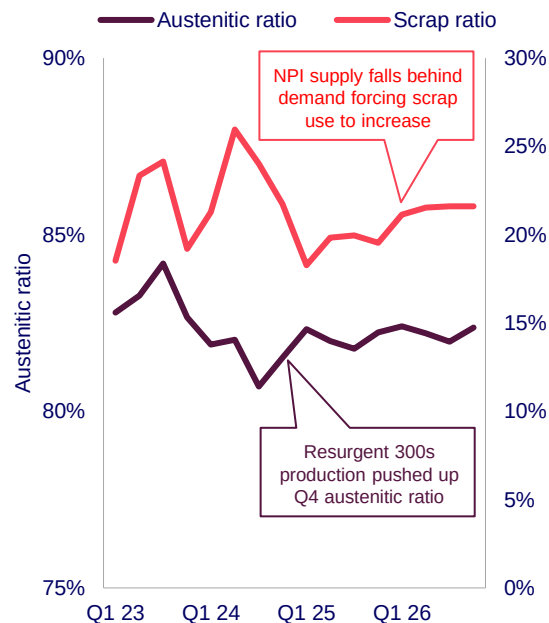
Chinese stainless production is at odds with demand: the former has increased considerably in 2024 while the latter has proven to be weaker than expected.

The rise in production is mostly because the two biggest producers – Tsingshan and Tisco (in Taiyuan, Desheng and Shandong Xinhai) - have increased output by ramping up spare capacity, apparently irrespective of the market.

Meanwhile, demand has disappointed in 2024 so the rise in output has led to oversupply. Ample spot supply has kept downward pressure on stainless prices – to a four-year low in December.

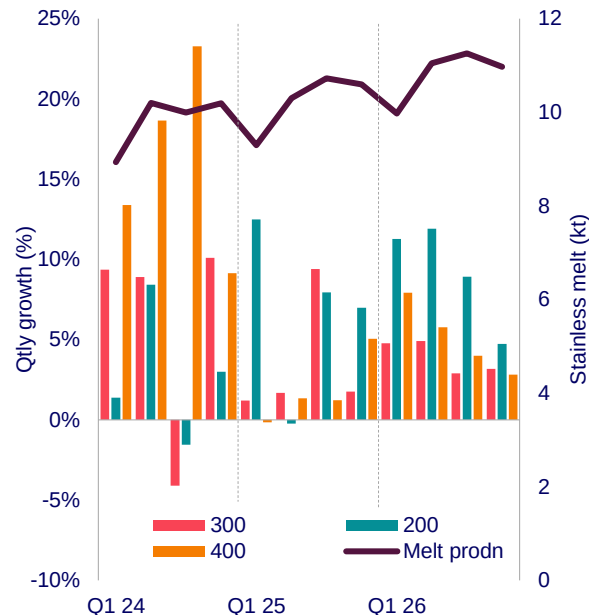
Falling prices have restrained buyers, who remain cautious, not least because of the need to de-stock by year end – typically a seasonal low - rather than buy new product.

Chinese austenitic and scrap ratios



Source: Wood Mackenzie

Changes in quarterly production (yoy)



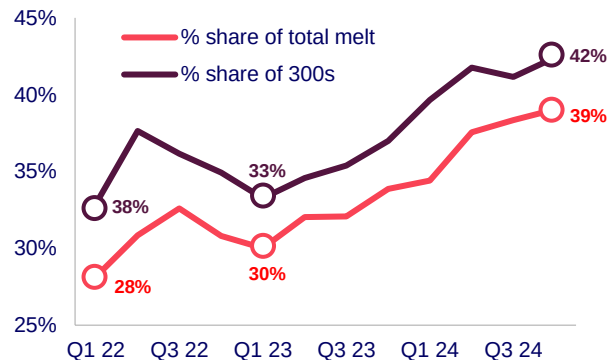
Source: Wood Mackenzie

Chinese stainless - the giants increase their market share aided by new capacity

Chinese stainless melt output remains steady with November production slightly below guidance and 1% lower than in October. This masks a net cut in 400s production while 300s output increased. Growth in the national total of 3% is targeted for December. This would leave production at 39.3 Mt in 2024.

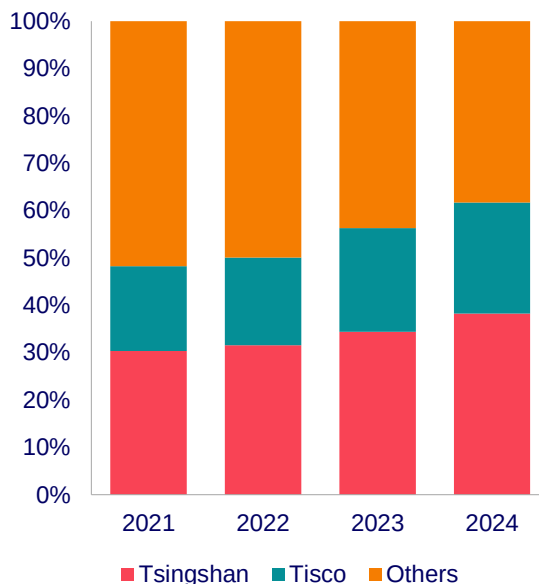
The increase in 300s in this period is mostly due to new capacity at Tsingshan's Fujian complex: a new hot strip annealing and pickling line; another new 1 Mtpa furnace. The latter is expected to feed a new plate line from later next year and, therefore, is perhaps being commissioned, with full commercial operations only coming once the plate line is ready.

Tsingshan's share of total and 300s melt



Source: Wood Mackenzie; ZLJ Steel

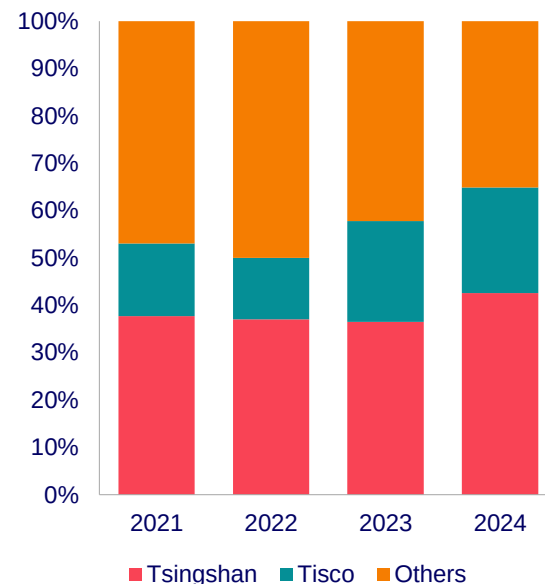
Total Chinese stainless melt production



Source: Wood Mackenzie; ZLJ Steel

Tsingshan has added at least 3 Mtpa of melting capacity at Fu'An in past 2-3 years. Together with Tisco, who added 1.5 Mtpa at Shandong Xinhai from late 2022, they will produce 62% of all Chinese stainless in 2024.

Chinese 300s stainless melt production



Source: Wood Mackenzie; ZLJ Steel

The two companies will also account for 65% of China's 300s this year, with Tsingshan alone increasing its share from 33% to 42% since the start of 2023.

European stainless market as gloomy as ever

Signs that market is worsening rather than getting better is undermining hopes for January and beyond

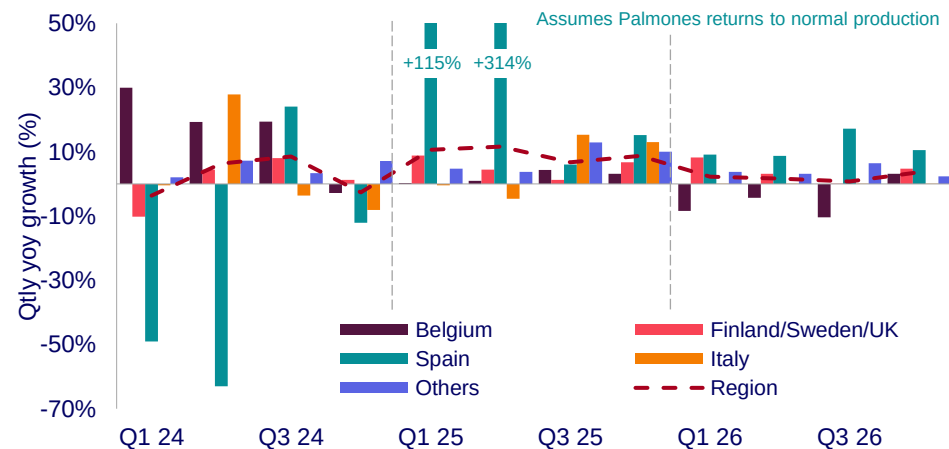
The European stainless steel market continues to grapple with significant challenges, with little indication of improvement in the near future. In recent months, there has been a notable escalation in electricity prices across key stainless steel-producing nations in Europe, with substantial rises in November being followed by further increases in December. In the major stainless producing countries of Belgium, Finland, Italy, and Spain, electricity costs in Q4 have surged by 63%, 52%, 10% and 19%, respectively, compared with Q3.

The primary drivers of these price hikes include colder weather, elevated natural gas prices, and reduced renewable electricity generation. Rising energy costs will not help margins that are already squeezed acutely and are likely part of the reason for the temporary production halts at Arvedi AST (Terni) and Acerinox in the past couple of months, especially with no improvement in stainless demand.

Indeed, on 12 December Outokumpu issued a profits warning that adjusted EBITDA in Q4 would “..be close to break even or turn negative..” due to the unexpectedly weak market and a maintenance break at Tornio that will now be longer than planned. Deliveries by the company could be as much as 10% down on Q3. Marcegaglia Sheffield (UK) also reportedly stopped production in early December and will not resume until 7 January 2025.

There are rumours that the mills are not entering contract negotiations for nickel

European stainless – growth in quarterly output, year-on-year



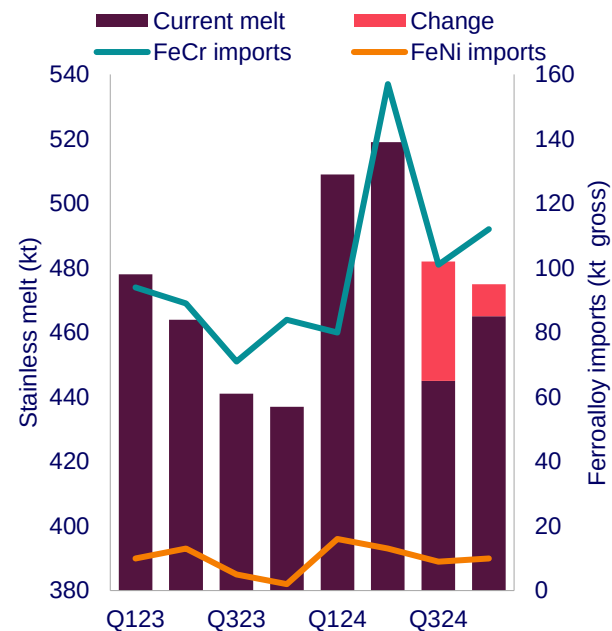
Source: Wood Mackenzie

raw materials in 2025, opting instead to buy from the spot market (which, incidentally, would probably include more NPI from Indonesia). This highlights how cautious and uncertain the mills are for next year. Production should pick up in January and then build through the following months; that would be normal, but the question, as always, is by how much? Demand needs to improve markedly, and distributors need to start re-stocking, but for that to happen, prices first need to turn upwards.

US stainless: gradual market improvement for most participants in 2025

But Crucible's potential insolvency by next March shows not all players will be around to benefit

US qtlly FeCr/FeNi imports & melt output



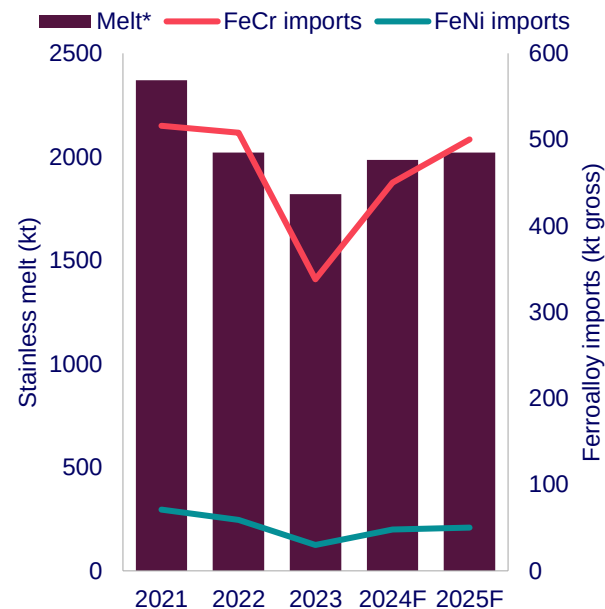
Source: Global Trade Tracker; USGS; Wood Mackenzie

In terms of stainless melt volume, 2024 is likely to end up being seen as a strong year for the US sector. Q3 data came in too late for us to reflect the 37 kt increase in our base case data, and adding a further 10 kt in for Q4 brings the year-end tally to 1.99 Mt, up by 9% on 2023. In the January Short-Term Outlook, we will also revise upwards our 2025 forecast by 120 kt, to 2.02 Mt – on a par with 2022.

Large-scale, scrap-intensive flat product makers dominate the stainless landscape. Long product makers usually melt a wide range of alloy, tool and stainless grades, but have seen more structural change of late. In October, Aperam bought Universal, and this month Crucible Specialty issued a notice to NY State authorities warning of entering Chapter 11 insolvency by next March.

While melters across the board should be able to raise output next year, raw material costs are also likely to rise – and for Class II nickel, availability could be an issue. FeCr supply seems adequate, but FeNi was largely sourced from just three countries in 2023 and 2024. From that list, Brazil and Colombia would appear to be the most reliable going into 2025, but risk still hangs over New Caledonia. One potential replacement and prior supplier might be Guatemala, which will re-start next year.

US annual FeCr/FeNi imports & melt output

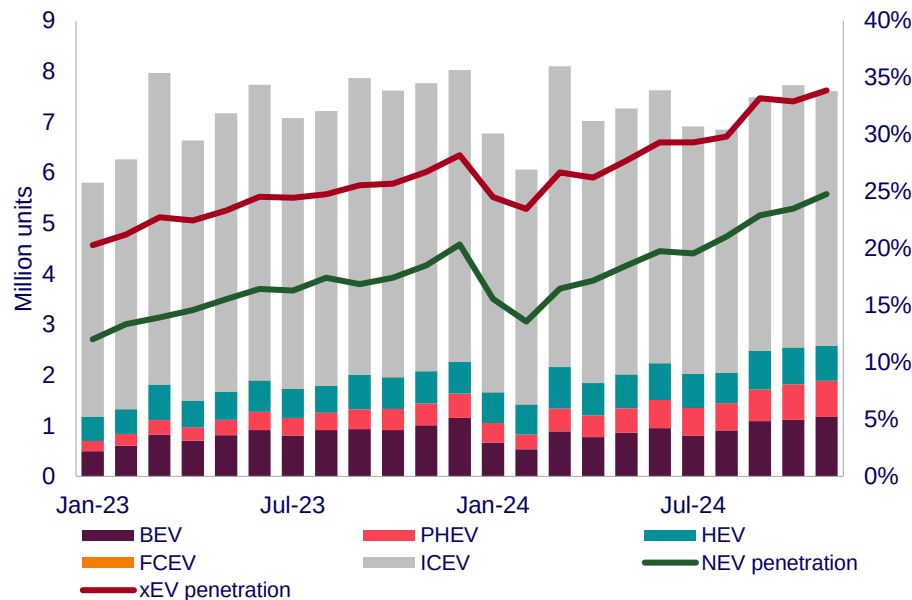


Source: Global Trade Tracker; USGS; Woodmac
*reflects revised forecasts

Global vehicle sales hit 7.6 million in November

YTD sales reach 67.6 million, 1% higher than the same period in 2023

Global vehicle sales and PEV penetration



China: The scrappage scheme continues to boost consumption levels with China recording 3.3 million passenger sales in November. Year-to-date (YTD) sales are at 23.48 million, 6% higher than the same period last year. On a YTD basis, while BEV penetration has risen marginally from 20% to 22% year on year, PHEV penetration has jumped from 9% to 16%, contributing heavily to the market share loss for ICEVs. The expected expiry of the scrappage scheme in December will keep the push on demand, with passenger sales set to exceed 26 million through 2024..

North America: Passenger sales reached 1.71 million units in November, 10% higher than 2023. YTD sales are at 18 million, 0.5 million higher than last year. On a YTD basis, while penetration of BEVs and PHEVs has stagnated at 7% and 1.5%, respectively, that of HEVs has risen from 6% to 8%, thanks to an aggressive sales push by OEMs like Toyota and Honda.

Rising days' supply across brands indicates December will see heavy discounting during the holiday period, likely boosting full year passenger sales to 20 million.

Europe: Europe continues to see poor consumption levels with, November sales declining 14% year on year. The drop in sales has been accompanied by changes in powertrain penetration. In Germany, while penetration of BEVs has dropped from 16% to 12% YTD, penetration of ICEVs has risen by 2%.

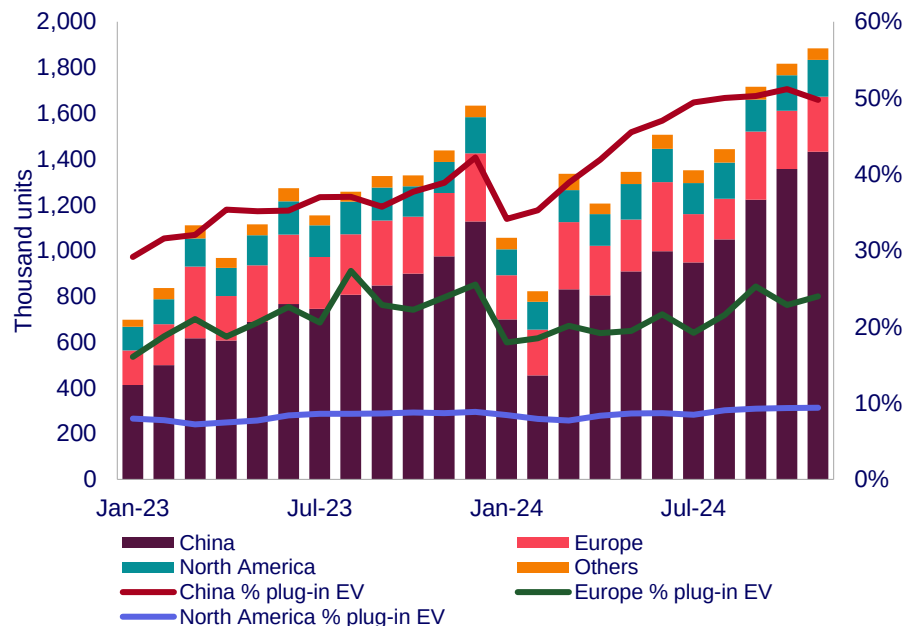
With little scope for recovery, full year sales are expected to remain under 14 million.

Source: Marklines, Wood Mackenzie. PEV = Battery-electric (BEV) + plug-in hybrid (PHEV), HEV = Hybrid electric, NEV = BEV + PHEV + fuel cell (FCEV), xEV = All powertrains except ICEV

Global PEV sales hit 1.9 million units in November

YTD PEV sales reach 15.46 million units, with full year sales expected to be less than 18 million

Sales of plug-in EVs by region



Source: Marklines, Wood Mackenzie, PEV = Battery-electric (BEV) + plug-in hybrid (PHEV)

China: Despite PEV penetration rate dropping marginally MoM, to 49.8%, sales hit a new record at 1.43 million units in November. YTD sales are at 10.7 million.

December sales will accelerate considerably with major OEMs offering heavy last-minute discounts and financing incentives to meet their 2024 sales targets. Tesla, Geely and Li Auto are now offering three-to-five-year 0% interest loans. Combined with the looming expiry of the US\$2,800 scrappage subsidy, the acceleration in demand is expected to push the country's EV sales to more than 12.2 million in 2024.

Europe: Issues with the German market continue to weigh on Europe with PEV sales dropping 13% year on year in November. YTD sales are at 2.59 million units.

While steep discounting in the UK in response to the ZEV mandate will boost country EV sales in December, in the EU, OEM pressure to sell off older ICEV models before the 2025 emission threshold kicks in suggests that the region's EV sales in 2024 will remain under 3 million, considerably lower than our Q4 forecast of 3.4 million.

North America: Regional EV sales reached a new record of 161,000 units in November, 19% higher than 2023. Year-to-date sales are at 1.56 million.

Sales in December are expected to accelerate with Tesla boosting incentives with free DC fast-charging and autonomy features with every purchase. Concerns around the federal PEV tax credit being eliminated in 2025 will keep demand, particularly for leased EVs, elevated. Despite expectations around growth, however, 2024 sales will likely remain under 1.85 million, lower than our Q4 forecast of 2.1 million.

Supply

Refined supply

Refined output to increase by 22% by 2026 compared with 2023

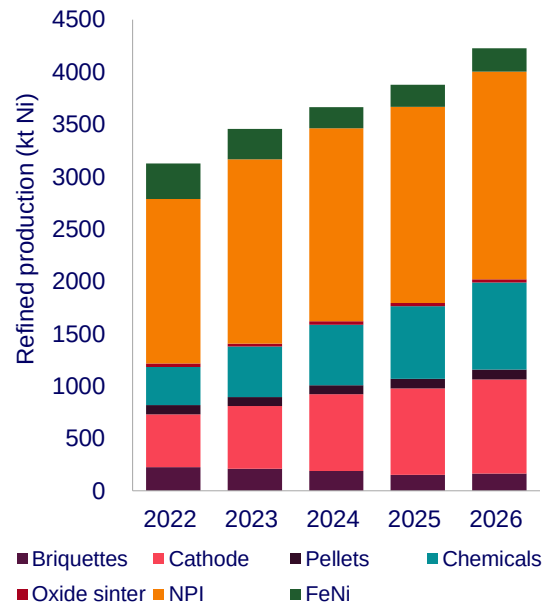
Class 1 output is forecast to increase by 29% by 2026, from 2023 levels, primarily from the refining of excess Chinese nickel sulphate to metal.

Chemicals output will increase by 73% over same period.

There is potential for swing production between sulphate and metal depending on relative prices and demand for sulphate.

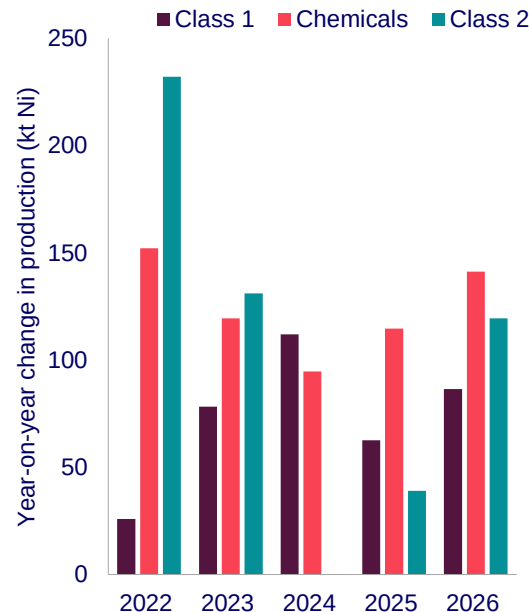
Growth in NPI/FeNi output has fallen significantly due to operational closures, project delays and anticipated slower new furnace line additions in Indonesia. Output is now forecast to rise by 8% by 2026. Overall, Class II production for 2024 is currently expected to be flat compared with 2023.

Refined production by type



Source : Wood Mackenzie

Growth in refined production by class



Source : Wood Mackenzie

Changes in forecast nickel production

Production in 2024 is lower by 132 kt compared with January forecast

Our forecasts for December are unchanged from the November figures.

Comparing our January 2024 data with December 2024 we have made cuts to our forecasts totaling 132 kt (3%), 228 kt (6%) and 107 kt (2%) for 2024 through to 2026, respectively.

The changes to our start of the year forecasts have predominantly come about mostly due to price pressure on western operations forcing closure. Civil unrest was a subsidiary issue in some locations.

Total cuts in Oceania production are 58 kt, 110 kt and 112 kt for 2024 through to 2026. Falling nickel prices decimated the Australian nickel industry and civil unrest in New Caledonia brought the mining industry to its knees.

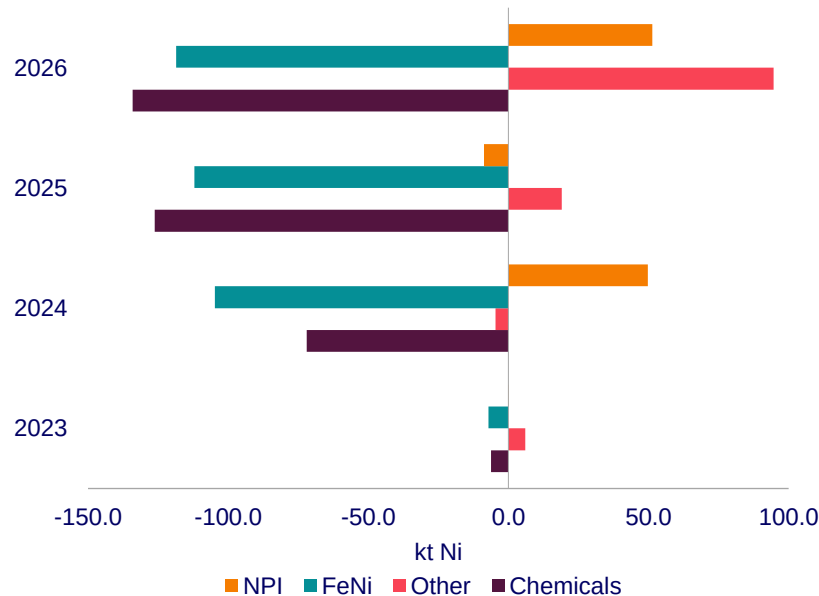
In Europe and Latin America low prices also closed FeNi operations. In Myanmar civil unrest placed Tagaung on care and maintenance.

Elsewhere, projects were cancelled or ran into financial difficulties (e.g. Araguaia in Brazil and Revolt Ett in Sweden).

Inevitably, Indonesian NPI output has increased from our initial forecasts, mainly due to Huabao starting all capacity and a switch back to NPI from matte production in Morowali.

Chemicals production is lower than originally forecast due to a larger swing to nickel cathode production as well as a deferred HPAL project in Indonesia (and the swing from matte to NPI lowering intermediate production).

Main changes in finished nickel output



Source: Wood Mackenzie

Matte and MHP payability forecasts

Intermediate payabilities should remain largely flat over the forecast period

Both matte and MHP payabilities were flat in December over November, averaging 85% and 82%, respectively. Our view is that payabilities will remain relatively flat over the forecast period, although averaging slightly lower than 2024.

	2023	2024	2025	2026
Matte	82%	84%	84%	83%
MHP	76%	80%	80%	79%

Source: Wood Mackenzie

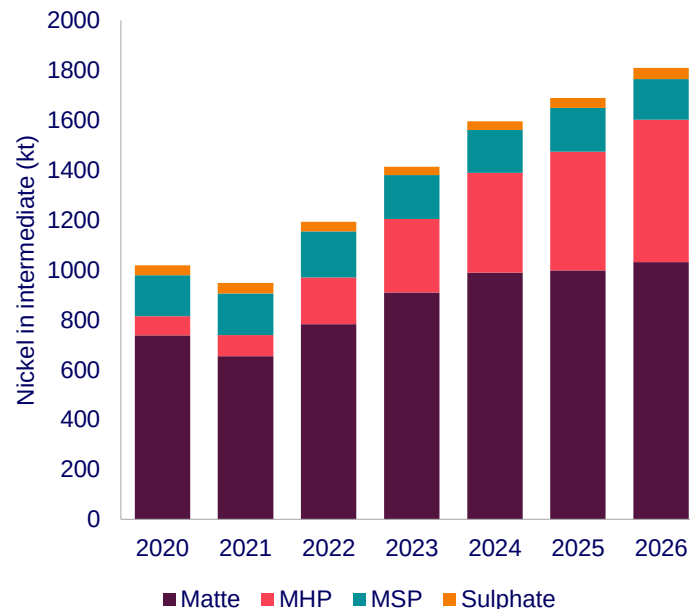
Payabilities are calculated based on current metal and sulphate prices, and conversion costs – providing an effective payability that can sustain a margin at a given price.

By 2026, nickel-in-matte production will increase by 13% over 2023 levels.

Nickel-in-MHP output will increase by 94% over the same time frame, solely through capacity additions in Indonesia.

There is considerable potential for the output of both MHP and matte to increase more strongly than shown, almost entirely from increases in Indonesian production.

Nickel-in-intermediates production



Source: Wood Mackenzie

Nickel sulphate production

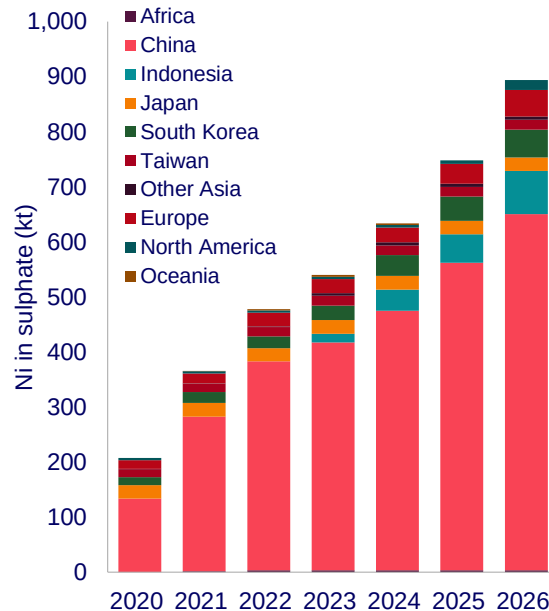
Nickel-in-chemicals production (predominantly as sulphate) will increase by 18% a year through to 2026

China currently accounts for 74% of global nickel sulphate production. This position is forecast to decline to 72% by 2026 on the assumption that new plants in Indonesia (e.g. Excelsior, which will have optionality for metal or MHP) produce sulphate.

Metal dissolution in China is now at minimal levels following the substantial increase in intermediates production in Indonesia.

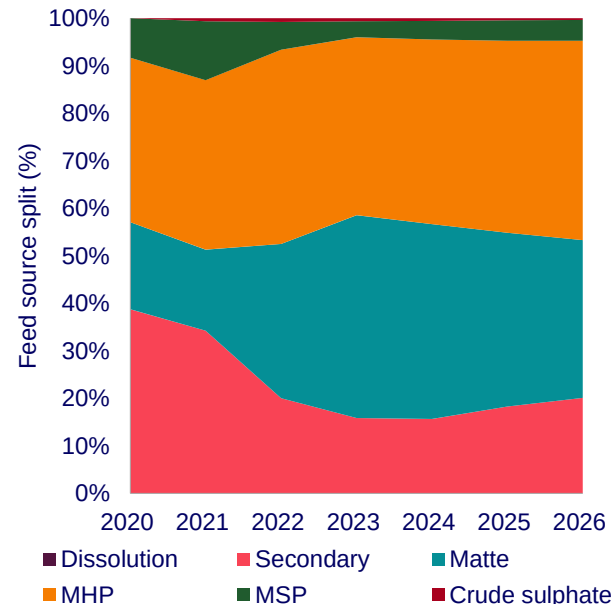
Potential exists to increase chemicals output should demand require it. But at present substantial tonnages of sulphate are being converted to metal.

Global nickel sulphate production



Source: Wood Mackenzie

Raw material feeds for NiSO_4



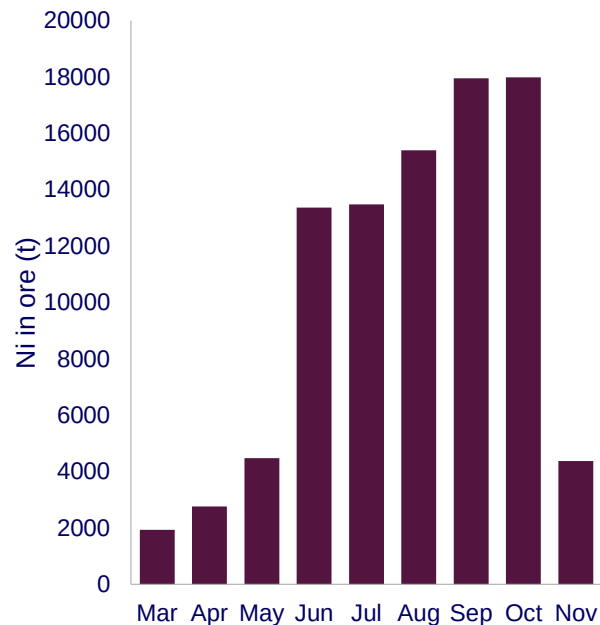
Source: Wood Mackenzie

Nickel supply analysis

Filipino ore imports into Indonesia

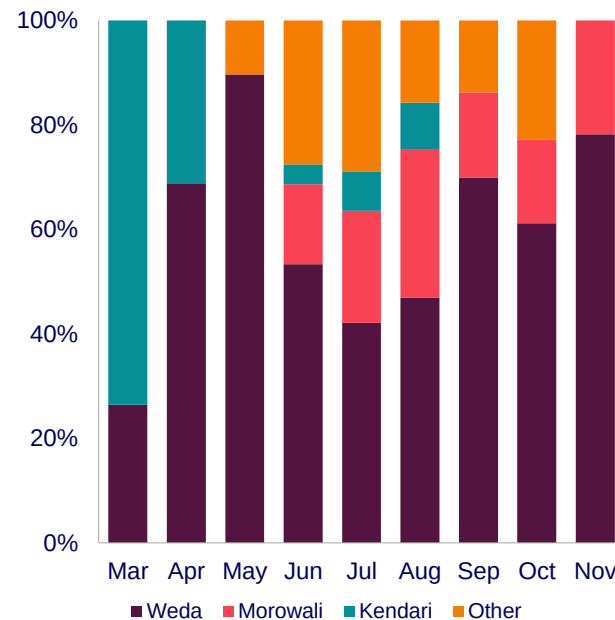
- Delays in ore mining permits and rising Indonesian ore prices resulted in Filipino ore imports starting in March.
- These increased through to October but have fallen sharply in November as the Philippines enters the monsoon season. We would expect December to be similar or lower than November.
- Nickel in ore for the nine months to September totalled 69 kt. We anticipate around 96 kt in total for 2024, which is about 4% of the required nickel-in-ore feed based on our total smelter output for the year.
- Around 60% of the ore imports have gone to Weda Bay, 20% to Morowali (IMIP) and 6% to Kendari (for Virtue Dragon and PT Gunbuster).
- We anticipate similar levels of imports to be supported in 2025 based on the ongoing discussions around RKAB permitted tonnages.

Monthly Filipino nickel in ore imports*



Source: Ferroalloy.net *Almost exclusively from Indonesia

Filipino ore imports by destination port



Source: Ferroalloy.net

Nickel supply analysis

US steps up funding for nickel projects and Portovesme hub studies are relaunched

Piaui project gets substantial funding promise. Brazilian Nickel announced it had received a Letter of Interest (LOI) from the US International Development Finance Corporation (DFC) for a US\$550 million development loan for its Piaui heap leach nickel project in Brazil. The DFC is also a shareholder in TechMet which owns 70% of Brazilian Nickel. The loan would meet 40% of the project's overall capital requirement for the proposed 27 ktpa nickel-in-MHP operation.

Piaui had previously received a US\$25 million investment from the US government in 2020 through TechMet, representing part of the previous Trump administration's efforts to reduce reliance of certain commodities on China.

Piaui established a pilot plant in 2022 (PNP1000), which began continuous production in July 2022 with a capacity of 1.4 ktpa. While challenges remain in securing the other US\$825 million required, such interest shown by the DFC will be supportive.

Talon Metals' US Department of Defence (DoD) aid. Talon Metals is looking to develop Tamarack deposit in Minnesota, which is currently under environmental review. The nickel from Tamarack is under offtake with Tesla. Our question has always been how the nickel concentrate from Tamarack would be processed to provide Tesla with the nickel it needs given there are no conventional sulphide concentrate processing facilities in the US.

The DoD contract for US\$2.47 million will be used to evaluate "novel" sulphuric acid and sodium hydroxide leaching as well as membrane electrodialysis at pilot plant scale over a 15-month period.

Portovesme metals hub gets a relaunch. Back in May 2023, Glencore and Li-Cycle announced they would jointly study the feasibility of developing a processing hub at Glencore's Portovesme site in Sardinia to produce nickel, cobalt and lithium from recycled materials. The project's future became questionable in October 2023 when the Sardinian regional government rejected a fast-track request. And now, in December 2024, Glencore has agreed to relaunch the study for the potential to develop a battery recycling hub in Portovesme.

The project was to be developed in two phases. The first would process 11 ktpa black mass, with phase 2 lifting capacity to 50-70 ktpa. The black mass would be sourced by Glencore and supplied by Li-Cycle's recycling facility located near Magdeburg, Germany.

Nickel supply analysis

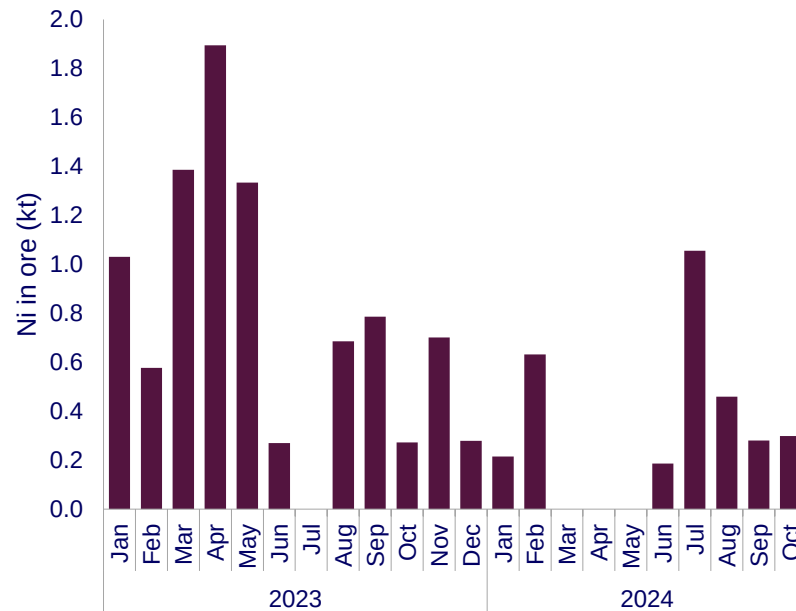
Guatemala resumes ore exports; FeNi production expected to re-start in 2025

We noted in our September outlook that sanctions on Compania Procesadora de Niquel de Izabal (PRONICO) and Compania Guatemalteca de Niquel (CGN) that had resulted in operations being suspended in March 2023 were removed in January of this year. It was unclear at that time whether ore export licences had been approved, although a licence to export FeNi was granted in August. Exports had also been hampered by low water levels in the Panama Canal and increased transit times.

By August this year, the traffic issue had largely subsided as booking slots were near normal daily averages (35 as of August 5, the pre-drought level being 36). Since then, existing ore stockpiles have been shipped; 75 kwmt had been stockpiled at the port prior to the cessation of operations. In addition, roughly one shipment per month has commenced to China. In its re-start plan, the company hoped to increase shipments to +100 kwmt a month. While still well below this level (currently ca. 25 kwmt a month), shipments are steady, and we expect tonnages to rise in 2025.

Production at Pronico is planned to re-start sometime in the second half of 2025.

Estimated Guatemalan nickel-in-ore exports



Source: Wood Mackenzie; Global Trade Tracker

Appendix

Key forecasts

Nickel key forecasts

	2023		2024		2025		2026	
Global (kt)								
Mine production	3812	+31.8%	3905	+2.4%	4016	+2.8%	4370	+8.8%
Smelter production	3475	31.4%	3647	4.9%	3716	1.9%	4004	7.8%
Refined production	3456	+27.2%	3657	+5.8%	3821	+4.5%	4170	+9.1%
Consumption	3170	+12.2%	3371	+6.4%	3720	+10.4%	3914	+5.2%
Market balance	286		286		100		256	
Cash LME price (US\$/t - current)	21,495		16,838		17,178		16,562	
(c/lb - current)	975		764		779		751	

Source: Wood Mackenzie

Nickel sulphate key forecasts

	2023		2024		2025		2026	
Global (kt Ni)								
NiSO4 production	540	+12.9%	633	+17.3%	765	+20.9%	917	+19.8%
Consumption in EV	491	+7.0%	519	+5.7%	639	+23.1%	746	+16.7%
Consumption in plating	38	+4.3%	39	+3.6%	40	+2.3%	41	+2.3%
Consumption in other	15	+0.8%	17	+7.5%	18	+6.1%	18	+1.6%
Total NiSO4 consumption	545	+6.6%	575	+34.7%	697	+6.6%	805	+5.6%
Market balance	-5		58		69		112	
NiSO4 price (US\$/t - current)	4,306		3,553		3,610		3,473	
(US\$/t Ni - current)	19,398		16,005		16,260		15,645	
Premium/discount to LME (US\$/t Ni = current)	-2,097		-833		-918		-917	

Source: Wood Mackenzie; the Argus Media group

Quarterly nickel consumption (kt)

Quarterly global nickel consumption in stainless steel

	2023 Year	2024					2025				
		Q1	Q2	Q3	Q4	Year	Q1	Q2	Q3	Q4	Year
SS output	59077	14967	15960	15797	16125	62849	15440	16875	16829	16952	66097
Aust output	47847	12047	12873	12594	13044	50557	12451	13594	13549	13725	53319
Scrap (%)	34	33	35	33	31	33	31	32	31	31	31
Aust (%)	81	80	81	80	81	80	81	81	81	81	81
Primary Ni	2033	536	535	535	570	2176	559	602	606	610	2377

Source: Wood Mackenzie

Quarterly refined nickel consumption by first use

	2023 Year	2024					2025				
		Q1	Q2	Q3	Q4	Year	Q1	Q2	Q3	Q4	Year
TOTAL	3170	831	835	833	872	3371	871	929	949	971	3720
Stainless steel	2033	536	535	535	570	2176	559	602	606	610	2377
Non-stainless	1136	295	300	298	302	1195	312	328	343	360	1344
Alloy steel	90	24	24	23	24	95	25	25	24	25	100
Non-ferrous alloys	241	62	63	62	64	252	66	66	66	67	266
Plating	171	44	44	44	45	177	45	45	45	46	182
Battery precursors	491	127	130	131	131	519	136	152	169	182	640
Foundry	66	17	17	17	18	69	18	18	17	18	70
Other	77	21	21	20	21	83	22	22	22	22	88

Source: Wood Mackenzie

Nickel stocks and premiums

	2023	2024 Q2	2024 Q3	2024 Jul	2024 Aug	2024 Sep	2024 Oct	2024 Nov	2024 Dec
Stocks - kt									
Antwerp	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Baltimore	1.5	0.6	0.4	0.5	0.5	0.4	0.4	0.3	0.3
Busan	8.3	18.8	25.1	21.7	23.1	25.1	25.9	29.0	28.9
Detroit	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Dubai	0.8	1.7	2.0	1.8	2.0	2.0	2.0	2.0	2.0
Gwangyang	1.6	4.8	9.7	7.8	9.8	9.7	9.7	9.6	8.3
Johor	2.9	5.2	5.2	5.2	5.2	5.2	8.7	11.7	8.4
Kaohsiung	9.4	11.4	25.4	17.7	21.8	25.4	34.0	39.0	39.7
Liverpool	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Mobile	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Port Klang	1.1	9.8	9.8	9.8	9.8	9.8	12.7	12.5	11.6
Rotterdam	25.9	26.3	30.9	26.3	26.1	30.9	30.8	31.0	31.4
Singapore (City)	12.7	16.9	22.7	19.1	21.0	22.7	22.5	28.1	31.1
Missingen	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Other	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total LME stocks	64.2	95.4	131.2	110.0	119.3	131.2	146.8	163.1	161.7
Cancelled warrants	6.3	4.9	8.1	6.5	8.9	8.1	8.1	11.3	14.4
Shanghai Futures Exchange	13.4	22.4	25.5	20.1	22.6	25.5	28.0	33.0	34.7
Prices									
LME									
Cash (US\$/t)	21464	18410	16255	16392	16246	16113	16800	15737	15678
3-month (US\$/t)	21690	18650	16512	16672	16488	16359	17064	16000	15918
Spread (3-month, cash)	227	240	257	280	242	246	264	263	240

	2023	2024 Q2	2024 Q3	2024 Jul	2024 Aug	2024 Sep	2024 Oct	2024 Nov	2024 Dec
SHFE									
Front month settlement RMB/t	169678	141962	129274	132096	129686	125381	129622	125750	126123
3-month settlement RMB/t	165842	142907	129922	132758	130301	126052	130535	126403	126940
Front month settlement US\$/t	21197	17346	15961	16096	16045	15682	16150	15446	15354
3-month settlement US\$/t	20718	17462	16041	16177	16121	15766	16264	15526	15454
Physical premiums (US\$/t)									
Europe plating	760	443	378	387	379	369	344	314	327
Europe melting	431	286	300	314	292	294	313	247	244
Europe FPC	369	222	197	205	194	193	170	146	153
Exchange Rates/US\$									
Australian \$	1.505	1.517	1.493	1.500	1.502	1.478	1.491	1.530	1.559
Canadian \$	1.350	1.369	1.364	1.373	1.365	1.355	1.373	1.397	1.412
Yen	140.49	155.71	149.22	157.74	146.23	143.22	149.63	154.04	151.18
Yuan	7.084	7.242	7.168	7.263	7.153	7.075	7.103	7.205	7.269
Euro	0.754	0.929	0.910	0.922	0.908	0.900	0.917	0.940	0.950

Source: LME, SHFE, IMF, the Argus Media Group, Wood Mackenzie

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